TECHNOLOGY STACK

GOOGLE MAPS BUSINESS ANALYSIS

Tools for collecting, cleaning, analyzing, and presenting local-business data.

Recommended workflow: Collect → Clean → Analyze → Visualize → Recommend

1. Technology Stack at a Glance

Layer	Tool(s)	Purpose
Data Collection	Google Maps public business listings + manual research	Gather business name, category, location, rating, review count, and available website link.
Data Handling	Microsoft Excel or Google Sheets	Store records, standardize fields, remove duplicates, handle missing values, and filter data.
Data Analysis	Excel formulas and PivotTables	Calculate counts and percentages; compare website presence by category and locality.
Analysis / Visualization	Power BI, Tableau, or Excel Charts	Build charts and a dashboard showing website status, category patterns, and priority opportunities.
Optional Advanced Analysis	Python: Pandas, Matplotlib, Seaborn	Automate cleaning, calculate summary statistics, and create repeatable charts for a larger dataset.
Presentation	Microsoft PowerPoint or Google Slides	Explain the method, key findings, visuals, top opportunities, and recommendations.

2. What Each Tool Does

Tool	Role in this project
Google Maps	Find local businesses and inspect public listing details. Follow platform terms; prefer manual or otherwise permitted collection.
Excel / Google Sheets	Main project workbook: raw data, cleaned data, formulas, filters, data validation, and summary tables.

Tool	Role in this project
Power BI / Tableau	Optional dashboard tools for interactive charts and category/location comparisons. Excel charts are enough for a simple submission.
Python + Pandas	Optional: load CSV/Excel files, standardize text, identify duplicates, handle missing values, and calculate website-status summaries.
Matplotlib / Seaborn	Optional Python visualization libraries for bar charts, doughnut/pie charts, and other summaries.
PowerPoint / Slides	Prepare the final presentation with project objective, methodology, results, charts, top-10 list, limitations, and recommendations.

3. Suggested Folder and File Structure

Google_Maps_Business_Analysis/
01_Raw_Data/ business_listings_raw.xlsx
02_Clean_Data/ business_analysis_clean.xlsx
03_Analysis/ summary_tables.xlsx
04_Charts_Dashboard/ charts_or_dashboard.pbix (if using Power BI)
05_Presentation/ project_insights.pptx
06_Final_Submission/ final_dataset.csv, final_report.pdf

4. Minimum Setup vs. Optional Advanced Setup

Setup	When to use it
Minimum (recommended for a beginner)	Google Maps + Excel/Google Sheets + Excel Charts + PowerPoint. This is sufficient for a clear, well-organized project.
Optional advanced workflow	CSV/Excel + Python (Pandas) for cleaning and analysis + Matplotlib/Seaborn for charts + Power BI/Tableau for a dashboard.

5. Workflow

Step	Action
1. Collect	Record business listing details and source reference.
2. Clean	Standardize names/categories, check duplicates, and label missing values.
3. Verify	Check for an official website; mark Found / Not Found / Unclear.
4. Analyze	Calculate totals, percentages, and category/location comparisons.
5. Visualize	Build charts or a dashboard with clear labels and dates.
6. Recommend	Prioritize businesses using transparent criteria and explain each suggestion.
7. Present	Submit the dataset and a PPT of findings and recommendations.

6. Important Notes

- Python, Pandas, Matplotlib, and Seaborn are optional, not mandatory, for this project.
- Do not treat a missing website field as proof that a business has no website. Verify and record uncertain cases.
- Keep the collection date and source/verification notes in the dataset.

- Follow Google Maps terms and applicable laws; do not use prohibited scraping or bypass restrictions.
- Choose one dashboard tool rather than trying to learn Power BI and Tableau both for a small project.

Recommended choice: Start with Excel or Google Sheets and Excel charts. Add Python or Power BI only if your dataset size, course rubric, or time allows it.